Ruiwen Wang

AI Systems | ML Systems | Distributed Training

Paris, France | (+33) 663113466
wangrw0124@gmail.com | French nationality
ruiwen.wang · Scholar · GitHub · LinkedIn

Ph.D. candidate (CIFRE) in Computer Science at Sorbonne University and EURECOM, working with Huawei Paris Research Center on efficient and reliable ultra-scale LLM and DNN training. Build symbolic performance and memory models with ILP and constraint-based planners for hybrid parallelism, pipeline scheduling, and recomputation under memory and communication constraints. Doctoral thesis unifies these into one portable planning workflow — predict, jointly optimise, re-plan, and adapt.

Large-scale LLM/DNN training • Symbolic performance and memory modeling • ILP / constraint optimization • Hybrid parallelism and pipeline scheduling

Validated on production Ascend clusters up to **16,384 NPUs** • Best reported speedup **36.72%** • 7 papers (4 first-author) • 3 patent families

Experience

Distributed and Parallel Computing Lab, Huawei Paris Research Center
Paris, France

Ph.D. Candidate (CIFRE) and Research Engineer – LLM Training Systems
2021 – 2026 (expected)

- Built cost-model-driven planners for hybrid and pipeline-parallel LLM training, covering stage partitioning, pipeline scheduling, and recomputation selection under memory and performance constraints.
- Designed symbolic peak-memory estimators, collective-cost models, and overlap-aware strategies for throughput and MFU tuning at large scale.
- Productionized research prototypes into deployable planning workflows with engineering and customer teams, delivering actionable configuration guidance, debuggable traces, and regression benchmarks across model families and scales.
- Supported technology transfer from research to production-scale training environments, including deployment feedback on customer workloads and long-sequence training scenarios.

Role progression: Research Apprentice (2021–2022) -> Research Engineer (2022–2023) -> CIFRE Ph.D. Candidate (2023–2026).

Selected Impact

- **BMPipe** (CLUSTER 2025): symbolic cost model plus ILP search for stage boundaries, chunking, and recomputation; achieved up to 1.36x speedup with 83–92% HBM utilization, validated on clusters up to 16,384 Ascend NPUs.
- **ManuMatic** (NPC 2025): treated operator shardings as first-class planning constraints; delivered 2.24x on Mixtral-8x7B and 2.04x on Llama3-8B versus D-Rec, plus 30.1% over an expert-tuned plan on Qwen2.5-72B.
- **H2O** (Euro-Par 2025 Best Poster): two-level workflow for parallelism degrees, micro-batch size, stage mapping, and adaptive recomputation; improved MFU from 26.13% to 35.73% and delivered 36.72% speedup on DeepSeek-141B, then transferred to production with about 1.1x on 4,096-device training and up to 2.37x on long-sequence training at 512 devices.
- **PRISM** (HPC Asia 2026): profiling-free symbolic planner for peak memory and communication; achieved 1.23x–1.43x speedups and raised MFU from 29.40% to 36.20%, validated up to 1,024 devices.

Core Areas

- Large-scale LLM and DNN training across tensor, data, pipeline, and expert parallelism
- Symbolic performance and peak-memory modeling for feasibility analysis
- ILP and constraint-based optimization for memory, communication, and scheduling trade-offs
- Pipeline scheduling, stage partitioning, micro-batch planning, and recomputation strategies
- Production-scale validation on GPU/NPU clusters and deployable benchmarking workflows
- Research-to-production transfer with engineering, platform, and customer teams

Selected Publications

- **BMPipe** – IEEE CLUSTER 2025
- **H2O** – Euro-Par 2025, Best Poster Award
- **ManuMatic** – NPC 2025
- **PRISM & MLLM Pipeline Bubble Modeling** – HPC Asia 2026
- **Concord**: dependency-keyed re-planning – under submission 2026
- **TeleChat3-MoE** training report – arXiv:2512.24157

Patents

Three patent families on execution-plan generation, load balancing, and multi-dimension parallelism configuration for large language model training.

Education

Sorbonne University & EURECOM 2023 – 2026 (expected)
Ph.D. in Computer Science — hybrid-parallel training optimisation; adv. Prof. Appuswamy & Prof. Li

Sorbonne University 2019 – 2022
M.Sc. in Computer Science

Paris-Saclay University 2016 – 2019
B.Sc. in Computer Science

Awards

- Golden Prize, Huawei Challenge Award for Automatic Load Balancing
- Best Poster Award, Euro-Par conference
- Letter of Appreciation, China Telecom

Skills

Programming: Python, C++, Java
ML frameworks: MindSpore, PyTorch
Platforms: large-scale GPU/NPU clusters and cloud
Tooling: Ascend profiling, tracing, performance regression benchmarking
Development: Linux, Git